Practical 1: Jira Setup & Space Creation

Jira is a powerful project management and issue-tracking tool used by teams to plan work, track progress, and manage projects.

Core Components of Jira

- **Issues:** The fundamental building blocks of work, categorized as Epics, Stories, Tasks, or Bugs.
- **Workflows:** The customizable set of stages (like *To Do*, *In Progress*, and *Done*) that an issue moves through.
- **Boards:** Visual displays of active work items mapped to specific workflow columns

A. Create a Jira Cloud Account and Explore the Dashboard

Step 1: Open Jira

1. Open your browser.
2. Go to Jira Cloud.
3. Click **Get it free** or **Get started**.
4. Sign in/create an Atlassian account.
5. Follow the setup instructions.

Step 2: Create your Jira site

Jira may ask you to choose a site name.

Step 3: Explore the dashboard

After logging in, explore the main navigation.

You may see options such as:

- **Home**
- **Spaces**
- **Plans**
- **Issues**
- **Boards**
- **Dashboards**
- **Reports**

The exact menu depends on your Jira plan/interface.

Dashboard: The dashboard gives you an overview of your Jira work.

You can see things such as:

- Assigned issues
- Recently viewed issues
- Charts
- Sprint information
- Filters
- Project/space information

B. Create a Company-Managed Scrum/Kanban Space

For your practical, you need a **company-managed** space.

Step 1: Go to Spaces

From Jira's main menu:

Spaces → Create space

Depending on your Jira interface, you may instead see:

Projects → Create project

Step 2: Select a template

You should see templates such as:

- Scrum
- Kanban
- Bug Tracking
- Project Management

For this practical, choose:

Scrum

if your assignment requires a Scrum space.

OR

Kanban

if your assignment requires a Kanban space.

Step 3: Select Company-managed

When Jira asks how you want to manage the space, select:

Company-managed

This is important because your later practicals require features such as:

- Workflows
- Components
- Versions
- Advanced configuration

Step 4: Enter space details

Example:

Space name:
Online Shopping System

Key:
OSS

You may also be asked to select an administrator.

Choose yourself if you are the administrator.

Click:

Create

C. Configure Space Details

Now open your newly created space.

Go to:

Space settings

You may find options such as:

- Details
- Access
- Work types
- People
- Components
- Versions
- Features
- Permissions
- Notifications

The exact options depend on your Jira setup.

Practical 2: Team Creation & Role Assignment

A. Add Team Members

The purpose of this practical is to add people who will work on your space.

Step 1: Open your Space

Go to:**Spaces → Your Space → Space settings**

Look for: **People** or**Access**or a **People** option in the space navigation.

Step 2: Add people

Click:

Add people

Enter the person's email address.

Select the appropriate access/role if Jira asks.

Click:

Add

The person should now appear as a member.

B. Assign Roles

A role determines what a person can do.

The exact role names available depend on your Jira setup.

Steps

1. Open **Space settings**.
2. Open **People/Access**.
3. Find the user.
4. Select/change their role.
5. Save the change.

Practical 3: Issue Types & Configuration

This practical has two parts:

A. Issue Types

B. Components and Versions

A. Create and Configure Issue Types

First understand the four important issue types.

1. Epic

An **Epic** is a large piece of work that can contain several smaller issues.

Example:

Epic: User Authentication

It can contain:

- Login
- Registration
- Forgot Password
- Logout

Think:

Epic = Big feature

2. Story

A **Story** represents a feature or requirement from the user's perspective.

3. Task

A **Task** is a specific piece of work.

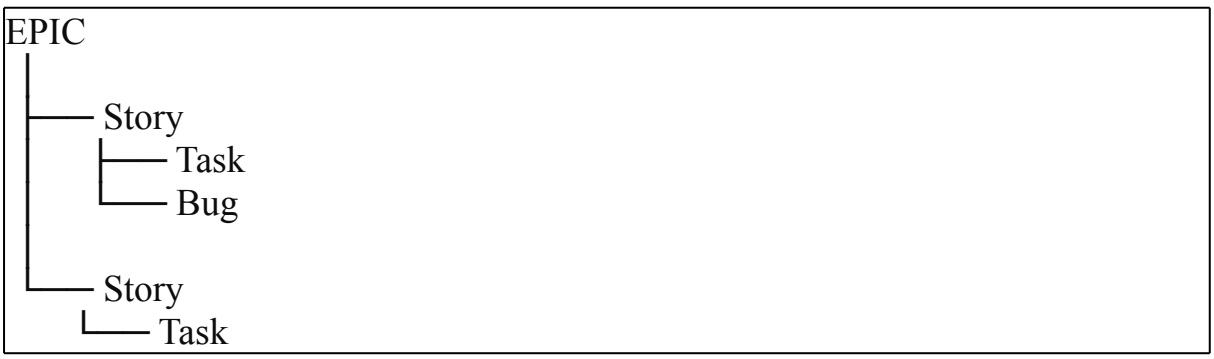
4. Bug

A **Bug** represents a problem/error in the application.

Bug = Problem/Error

Issue hierarchy

For easy understanding:



The exact hierarchy and parent/child relationships can vary by Jira configuration, so use the fields Jira provides in your instance.

How to Configure Issue Types

For a company-managed space, issue-type configuration is generally handled through Jira administration.

Depending on your permissions, you may access:

Settings → Issues → Issue types

or the relevant **Work types/Issue types** configuration area.

To add an issue type

1. Open Jira **Settings**.
2. Go to the issue/work-type configuration area.
3. Select **Add issue type**.
4. Enter the name.
5. Choose the appropriate type/category if Jira asks.
6. Save.

For your practical, make sure you have access to:

- Epic
- Story
- Task
- Bug

If these already exist, **you do not need to create duplicates**. You can simply configure/use them.

Create an Epic

Go to:

Create

Select:

Issue type → Epic

Example:

Summary:

User Authentication

Click:

Create

Create a Story

Click:

Create

Select:

Story

Create a Task

Example:

Task:

Design Login Page

Set its parent appropriately if your configuration provides a parent field.

Create a Bug

Example:

Bug:

Login button not responding

Again, select the appropriate parent if available.

B. Create Components and Versions

What is a Component?

A **Component** is a part/module of your application.

For an online shopping application, components could be:

- Login
- Payment
- Shopping Cart
- Product Catalog
- Database

Think:

Component = Part of the system

Create Components

Go to:

Space settings → Components

Click:

Create component

Enter:

Name:

Login Module

You can also enter:

- Description
- Component lead
- Default assignee

Click:

Create

Repeat for:

- Payment Module
- Shopping Cart
- Product Catalog

What is a Version?

A **Version** represents a release of your software.

For example:

- Version 1.0
- Version 1.1

- Version 2.0

Think:

Version = Release of the software

Create a Version

Go to:

Space settings → Versions

Click:

Create version Enter:**Name:**Version 1.0You can optionally enter:**Start date**

Release date

Description

Example:

Initial release of the online shopping application.

Click **Create**.

Practical 4: Issue Management

A. Create Issues Under an Epic

Let's use this example:

Epic

Online Shopping System

Under this Epic, create:

Story 1

User Registration

Story 2

User Login

Task

Design Registration Page

Bug

Registration button not working

Step 1: Create the Epic

Click:

Create

Select:

Epic

Enter:

Summary:

Online Shopping System

Click:

Create

Step 2: Create a Story under the Epic

Click:

Create

Select:

Story

Enter:

User Registration

Find the **Parent** field if available.

Select:

Online Shopping System

Click:

Create

Step 3: Create another Story

Create:

User Login

Set its parent to:

Online Shopping System

Step 4: Create a Task

Create:

Design Registration Page

Set its parent appropriately according to your Jira configuration.

Step 5: Create a Bug

Create:

Registration button not working

Set its parent appropriately if Jira provides the relevant parent field.

B. Assign Issues to Users

Now we'll assign the issues to team members.

Open an issue.

Find:

Assignee

Click it.

Select a team member.

For example:

User Registration → Rahul

User Login → Amit

Design Registration Page → Priya

Registration button not working → Rahul

Click **Save** if necessary.

Set Priority

Open an issue.

Find:

Priority

Select something like:

- Highest
- High
- Medium
- Low
- Lowest

Example:

Registration button not working

Set:

Priority → Highest

Because it is an important defect for your practical example.

User Login

Set:

Priority → High

Set Due Date

Open the issue.

Find:

Due date

Choose a date.

Practical 5: Workflow Customization

A. Design a Custom Workflow

What is a workflow?

A **workflow** is the path an issue follows from beginning to end. A Jira workflow is the specific sequence of statuses and transitions a task or issue moves through from creation to completion

Core Elements of a Jira Workflow

- **Status**: Represents the state of a work item at any given time (e.g., *To Do*, *In Progress*, *Done*).
- **Transition**: The directional link or action that moves an issue from one status to another.
- **Workflow Scheme**: Maps specific workflows to particular issue types within a project

For example:

To Do → In Progress → Testing → Done

Here:

- **To Do** = issue has not started
- **In Progress** = work is being done
- **Testing** = work is being tested
- **Done** = work is completed

Steps

1. Open your Jira project.
2. Go to **Project settings**.
3. Find **Workflows**.
4. Create or edit a workflow.
5. Add statuses such as:
    - To Do
    - In Progress
    - Testing
    - Done
6. Arrange them in the required order.

B. Add Transitions and Conditions

What is a transition?

A **transition** is the movement from one status to another.

Example:

To Do → In Progress

The transition could be called **Start Work**.

Another:

In Progress → Done

The transition could be called **Complete Work**.

What is a condition?

A **condition** controls who can perform a transition or when it can happen.

C. Publish and Associate Workflow

After designing the workflow:

1. Click **Publish**.
2. Select the project/issue type where you want to use it.
3. Associate the workflow with the project.
4. Save/apply the changes.

Practical 6: Scrum Board & Sprint Management

A. Create and Configure Scrum Board

What is Scrum?

Scrum is an Agile method where work is divided into short periods called **Sprints**.

A **Scrum Board** usually contains:

To Do → In Progress → Done

Steps

1. Go to **Boards**.
2. Select **Create board**.
3. Choose **Scrum board**.
4. Select your project or create a new project.
5. Give the board a name.
6. Configure the columns.

Example:

To Do	In Progress	Done
Login page	Database	Homepage
Registration	API	Navbar

B. Create and Start a Sprint

What is a Sprint?

A **Sprint** is a fixed period during which the team works on selected issues.

For example:

Sprint 1 = 2 weeks

Steps

1. Open your Scrum Board.
2. Go to **Backlog**.
3. Click **Create sprint**.
4. Select issues from the backlog.
5. Drag the issues into the sprint.
6. Click **Start sprint**.
7. Set:

    - Sprint name
    - Start date
    - End date
8. Start the sprint.

Example:

Sprint 1

- Create Login
- Create Registration
- Create Dashboard

C. Complete Sprint and Generate Sprint Report

When the sprint is finished:

1. Open the active sprint.
2. Move completed issues to **Done**.
3. Click **Complete sprint**.
4. Jira will show the sprint summary/report.
5. Open **Reports**.
6. Select **Sprint Report**.

The Sprint Report helps you see:

- Issues completed
- Issues not completed
- Work added/removed
- Sprint progress

Easy way to remember

Backlog → Create Sprint → Add Issues → Start Sprint → Work → Complete Sprint → Sprint Report

Practical 7: Kanban Board Management

A. Create a Company-Managed Kanban Project

What is Kanban?

Kanban is a method of managing work by **visualizing tasks and continuously moving them through different stages**.

Unlike Scrum, Kanban normally does **not require fixed-length sprints**.

Example:

To Do → In Progress → Done

Steps

1. Go to **Projects**.
2. Select **Create project**.
3. Select **Kanban**.
4. Choose **Company-managed project**.
5. Enter the project name.
6. Create the project.

B. Configure Kanban Board

Open the Kanban board.

You can configure columns such as:

To Do	In Progress	Testing	Done
Task 1	Task 2	Task 3	Task 4

You can also configure:

- Columns
- Statuses
- Quick filters
- Card information
- Work-in-progress limits

What is WIP?

WIP = Work In Progress

It limits how many issues can be in a particular stage.

For example:

In Progress WIP limit = 3

This means the team should have no more than 3 issues in that column at once.

C. Track Issue Progress and Generate Control Chart

Move issues across the board:

To Do → In Progress → Testing → Done

Jira records how long issues take to move through the workflow.

Control Chart

The **Control Chart** shows how long issues take to complete.

It helps you understand:

- How quickly issues are completed
- How long issues stay in progress
- Whether completion time is increasing/decreasing
- Possible bottlenecks

Steps

1. Open your Kanban project.
2. Go to **Reports**.
3. Select **Control Chart**.
4. Select the required time period/filter.
5. Analyze the chart

Practical 8: JQL & Filters

This one is very important because **JQL is basically Jira's search language**.

What is JQL?

JQL = Jira Query Language

It allows you to search for specific issues.

Think of it like asking Jira:

"Show me all issues that are currently In Progress."

A. Basic JQL Queries

Go to:

Issues → Search for issues Then switch to **Advanced/JQL** mode.

1. Search by Status

Staus:"In Progress"

Meaning:

Show all issues whose status is In Progress.

Another example:

Staus:"Done"

Shows completed issues.

2. Search by Assignee

Assigned:User

Show issues assigned to that user.

The exact account identifier/name available in your Jira instance may differ.

3. Search by Priority

Priority=”High”

Meaning:

Show all High-priority issues.

Combine Conditions

You can combine conditions using **AND**.

Example:

```
status = "In Progress" AND priority = High
Done
```

Show completed issues assigned to John.

B. Create and Save Filters

A **filter** is a saved JQL search.

For example, suppose you frequently need to see:

High-priority issues that are currently In Progress.

Write:

```
status = "In Progress" AND priority = High
```

Then:

1. Run the search.
2. Click **Save as**.
3. Give the filter a name.
4. Example: **High Priority In Progress Issues**
5. Save it.

Now you can open the saved filter whenever you need it instead of typing the JQL again.

Practical 9: Reports and Dashboards

A.**Generate Velocity Chart**

1. Open the project in Jira.
2. Navigate to the **Reports** section. Click on More Reports
3. Select **Velocity Chart**.
4. Select the required number of sprints.
5. Observe the committed and completed work for each sprint.
6. Compare the team's velocity across the sprints.
7. Capture the generated chart.

B. Create Pie Chart

1. Open the project dashboard/report section.
2. Create or select an issue filter containing the required project issues.
3. Select the **Pie Chart** report/gadget.
4. Select a field such as **Issue Type** or **Priority**.
5. Generate the chart.
6. Observe the distribution of issues among the selected categories.
7. Capture the generated Pie Chart.

Generate Issue Statistics Report

1. Open the project reports section.
2. Select **Issue Statistics**.
3. Select the required issue filter.
4. Select the field by which the issues should be grouped, such as Status, Priority, or Issue Type.
5. Generate the report.
6. Analyse the number of issues in each category.

7. Record the observations.